Multi-Patch Global-to-Local Transformer Architecture for Efficient Flow Matching and Diffusion Model

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Introduction

- DiT provide strong generation quality but are expensive as every block processes the same tokens.
- Standard DiT uses a fixed patch size and pays the attention cost throughout the full network.
- MPDiT reduces computation while preserving high-fidelity class-conditional generation.

Key ideas

Global-to-local transformer:

early blocks use large patches for coarse global context; later blocks use smaller patches for local refinement.

Multi-patch hierarchy:

patch sizes {4, 2}; first N - k blocks operate on 64 tokens, final k blocks operate on 256 tokens.

FNO time embedding

improves temporal representation and training convergence.

Multi-token class conditioning

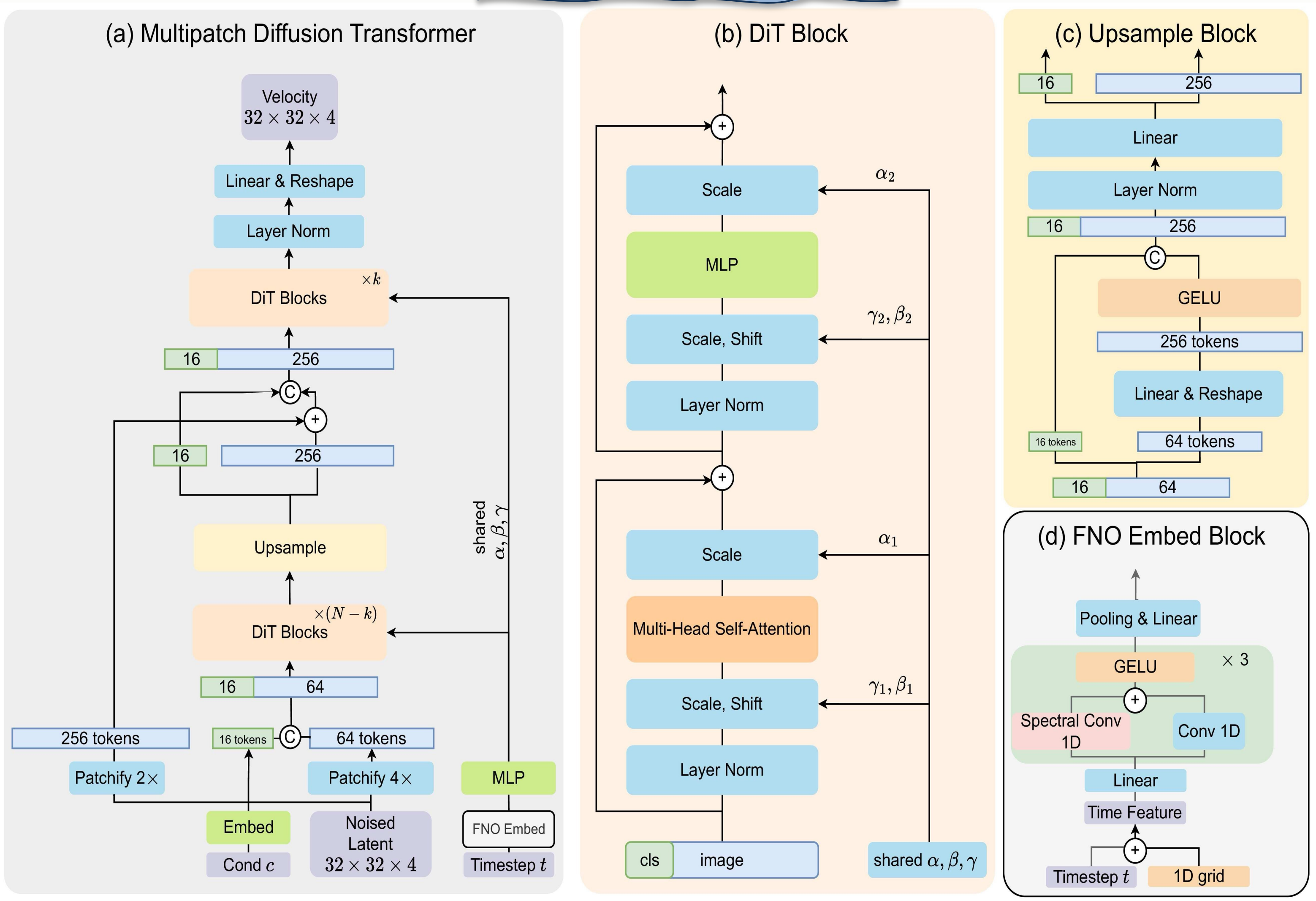
provides richer label supervision and faster convergence.

Up to 50% lower GFLOPs than standard DiT while maintaining strong generation quality.

Model	N	k	Model Dim D	GFLOPs ↓	$\frac{\text{GFLOPs}_{\text{MPDiT}}}{\text{GFLOPs}_{\text{DiT}}}$ ↓
MPDiT-B	12	6	768	16.6	72.1%
MPDiT-XL	28	6	1152	59.3	49.9%

Design intuition: spend most blocks on coarse global tokens, then reserve only a few blocks for fine local synthesis.

MPDiT architecture



- Early blocks capture global context with only 64 tokens
- Upsample expands 64 -> 256 tokens.
- Only the last k blocks operate at fine resolution.
- Shared condition is used across DiT blocks.

Qualitative results

ImageNet-256 samples (256x256)

ImageNet-512 samples (512x512)

Experiments

Model	Epochs	GFLOPs↓	FID↓	sFID↓	IS↑	Pre↑	Rec↑
ADM [16]	–	–	10.94	6.02	100.98	0.69	0.63
ADM-U	–	–	7.49	5.13	127.49	0.72	0.63
ADM-G, ADM-U	–	–	3.94	6.14	215.84	0.83	0.53
CDM	–	–	4.88	–	158.71	–	–
LDM-8 [52]	–	–	15.51	–	79.03	0.65	0.63
LDM-8-G	–	–	7.76	–	209.52	0.84	0.35
LDM-4-G	–	–	3.60	–	247.67	**0.87**	0.48
Mamba and State Space Diffusion Model							
DiM-H/2 [60]	–	–	2.40	–	–	–	–
DIFFUSSM-XL [72]	515	–	2.28	4.49	259.13	0.86	0.56
DiMSUM-L/2-G [50]	510	–	2.11	–	–	–	0.59
Our Model and Baselines							
SiT-B/2 [40]	80	23.02	34.84	6.59	41.53	0.52	0.64
DiT-B/2 [48]	80	23.02	43.47	–	–	–	–
DiG-B/2 [84]	80	17.07	39.50	8.50	37.21	–	–
DiC-B [63]	80	23.50	32.33	–	48.72	0.51	0.63
DiCo-B [1]	80	16.88	27.20	–	56.52	**0.60**	0.61
MPDiT-B (ours)	80	**16.60**	**24.74**	**6.32**	**57.40**	0.58	**0.65**
SiT-XL/2 [40]	80	118.66	18.04	5.07	73.90	0.63	0.64
DiT-XL/2 [48]	80	118.66	19.47	–	–	–	–
DiG-XL/2 [84]	80	89.40	18.53	6.06	68.53	0.63	0.64
DiC-XL [63]	80	116.1	13.11	–	100.15	–	–
DiCo-XL [1]	80	87.30	11.67	–	100.42	**0.71**	0.61
MPDiT-XL (ours)	80	**59.30**	**9.92**	**5.05**	**102.79**	0.70	**0.64**
SiT-XL/2 [40]	1400	118.66	9.35	6.38	126.06	0.67	0.68
DiT-XL/2 [48]	1400	118.66	9.62	6.85	121.50	0.67	0.67
DiG-XL/2 [84]	240	89.40	8.60	6.46	**130.03**	0.68	0.68
MPDiT-XL (ours)	240	**59.30**	**7.36**	**5.09**	127.57	**0.69**	**0.69**
SiT-XL/2-G [40]	1400	118.66	2.15	4.60	258.09	0.81	0.60
DiT-XL/2-G [48]	1400	118.66	2.27	4.60	278.24	**0.83**	0.57
DiG-XL/2-G [84]	240	89.40	2.07	4.53	**278.95**	0.82	0.60
MPDiT-XL-G (ours)	240	**59.30**	**2.05**	**4.51**	278.73	0.82	**0.61**

Table 1. Quantitative performance of MPDiT on ImageNet 256.

Model	Params(M)	GFLOPs	FID ↓	Prec ↑	Rec ↑
BigGAN [3]	160	-	8.43	**0.88**	0.29
StyleGAN-XL [56]	166	-	2.41	0.77	0.52
MaskGIT [5]	227	-	7.32	0.78	0.50
VQ-GAN [18]	227	-	26.52	0.73	0.31
VAR-d36-s [61]	2300	-	2.63	-	-
ADM-U [16]	731	2813	3.85	0.84	0.53
U-ViT-L/4 [2]	287	76.5	4.67	0.87	0.45
U-ViT-H/4 [2]	501	133.3	4.03	0.84	0.48
Simple Diff [29]	2000	-	4.53	-	-
VDM++ [32]	2000	-	2.65	-	-
DiT-XL/2 [48]	675	524.7	3.04	0.84	0.54
SiT-XL [40]	675	524.7	2.62	0.84	**0.57**
DiM-H [60]	860	708	3.78	-	-
DiffuSSM-XL [72]	673	1066.2	3.41	0.85	0.49
DiCo-XL [1]	701	349.8	2.53	0.83	0.56
MPDiT-XL (ours)	482	228.4	**2.47**	0.83	0.56

Table 8. Quantitative results of ImageNet 512 with $\text{MPDiT}_{k=6}$

Method	Epoch	Params(M)	GFLOPs↓	FID↓
No cfg				
DiT-XL/2	600	675	524.7	11.93
$\text{MPDiT}_{\{22,6\}}$	120	482	228.4	9.24
$\text{MPDiT}_{\{18,6,4\}}$	120	491	138.2	11.77
With cfg				
DiT-XL/2	600	675	524.7	3.04
SiT-XL/2	600	675	524.7	2.62
$\text{MPDiT}_{\{22,6\}}$	120	482	228.4	2.47
$\text{MPDiT}_{\{18,6,4\}}$	120	491	138.2	3.13

Table 9. Performace of different MPDiT-XL variants on ImageNet 512

Ablation

Method	Params(M)	GFLOPs↓	FID↓
Configuration B			
DiT-B/2 †	101.2	24.3	24.52
MPDiT $k = 4$	104.8	13.9	26.94
MPDiT $k = 6$ ***(default)***	104.8	16.6	24.74
MPDiT $k = 8$	104.8	19.3	24.62
Configuration XL			
DiT-XL/2 †	473.1	125.5	9.22
MPDiT $k = 4$	481.2	53.2	11.11
MPDiT $k = 6$ ***(default)***	481.2	59.3	9.92
MPDiT $k = 8$	481.2	65.4	9.73

Table 4. Ablation on k value for MPDiT. † means that DiT with Shared AdaIN, multitokens class and FNO time embedding.

Method	Params(M)	GFLOPs↓	FID↓
Traditional Time Embedding			
`1×Linear`	104.9	16.6	28.98
`2×Linear` ***(default)***	105.5	16.6	27.92
`3×Linear`	106.1	16.6	29.71
FNO Time Embedding			
`2×MixedFNO`	104.8	16.6	26.87
`3×MixedFNO` ***(default)***	104.8	16.6	24.74
`4×MixedFNO`	104.8	16.6	27.37

Table 5. Ablation on the time embedding module. We vary the number of `Linear` layers and `MixedFNO` blocks. The traditional time embedding applies two `Linear` layers to sinusoidal time features, whereas the proposed FNO time embedding uses three `MixedFNO` blocks operating on grid-based time features.

Method	Params(M)	GFLOPs↓	FID↓
$m = 1$	93.3	15.3	32.31
$m = 4$	95.6	15.6	30.91
$m = 8$	98.7	15.9	28.12
$m = 16$ ***(default)***	104.8	16.6	24.74
$m = 32$	117.1	18.0	24.47

Table 6. Ablation on number of class tokens m

Method	Params(M)	GFLOPs↓	FID↓
`Linear`	104.2	16.5	27.06
`ConvTranpose`	104.2	16.9	29.45
`Linear+Linear` ***(default)***	104.8	16.6	24.74
`Linear+MLP(r=4)`	109.0	17.8	24.85
`Linear+Conv`	104.8	16.6	25.68

Table 7. Ablation on design of Upsample Block. r is the mlp ratio.